

Software Project Scheduling

- Introduction
- Project scheduling
- Task network
- Timeline chart
- Earned value analysis

Project Scheduling

General Practices

- On large projects, hundreds of small tasks must occur to accomplish a larger goal
 - Some of these tasks lie outside the mainstream and may be completed without worry of impacting on the project completion date
 - Other tasks lie on the critical path; if these tasks fall behind schedule, the completion date of the entire project is put into jeopardy

General Practices

- Project manager's objectives
 - Define all project tasks
 - Build an activity network that depicts their interdependencies
 - Identify the tasks that are critical within the activity network
 - Build a timeline depicting the planned and actual progress of each task
 - Track task progress to ensure that delay is recognized "one day at a time"
 - To do this, the schedule should allow progress to be monitored and the project to be controlled

Relationship Between People and Effort

- Common management myth: *If we fall behind schedule, we can always add more programmers and catch up later in the project*
 - This practice actually has a disruptive effect and causes the schedule to slip even further
 - **The added people must learn the system**
 - The people who teach them are the same people who were doing the work earlier
 - During teaching, no work is being accomplished
 - **Lines of communication (and the inherent delays) increase for each new person added**

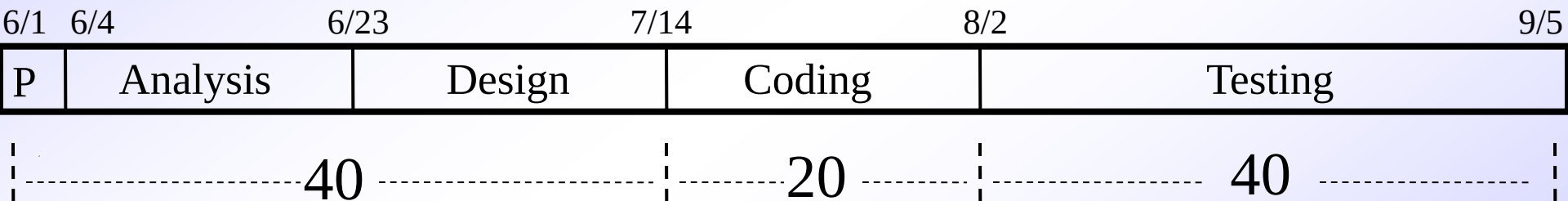
40-20-40 Distribution of Effort

- A recommended distribution of effort across the software process is 40% (analysis and design), 20% (coding), and 40% (testing)
- Work expended on project planning rarely accounts for more than 2 - 3% of the total effort
- Requirements analysis may comprise 10 - 25%
 - **Effort spent on prototyping and project complexity may increase this**
- Software design normally needs 20 – 25%
- Coding should require only 15 - 20% based on the effort applied to software design
- Testing and subsequent debugging can account for 30 - 40%
 - Safety or security-related software requires more time for testing

(More on next slide)

40-20-40 Distribution of Effort (continued)

Example: 100-day project



Task Network

Defining a Task Set

- A task set is the work breakdown structure for the project
- No single task set is appropriate for all projects and process models
 - It varies depending on the project type and the degree of rigor (based on influential factors) with which the team plans to work
- The task set should provide enough discipline to achieve high software quality
 - But it must not burden the project team with unnecessary work

Factors that Influence a Project's Schedule

- **Size of the project**
- **Number of potential users**
- Mission criticality
- Application longevity
- Stability of requirements
- Ease of customer/developer communication
- Maturity of applicable technology
- Performance constraints
- Embedded and non-embedded characteristics
- Project staff
- Reengineering factors

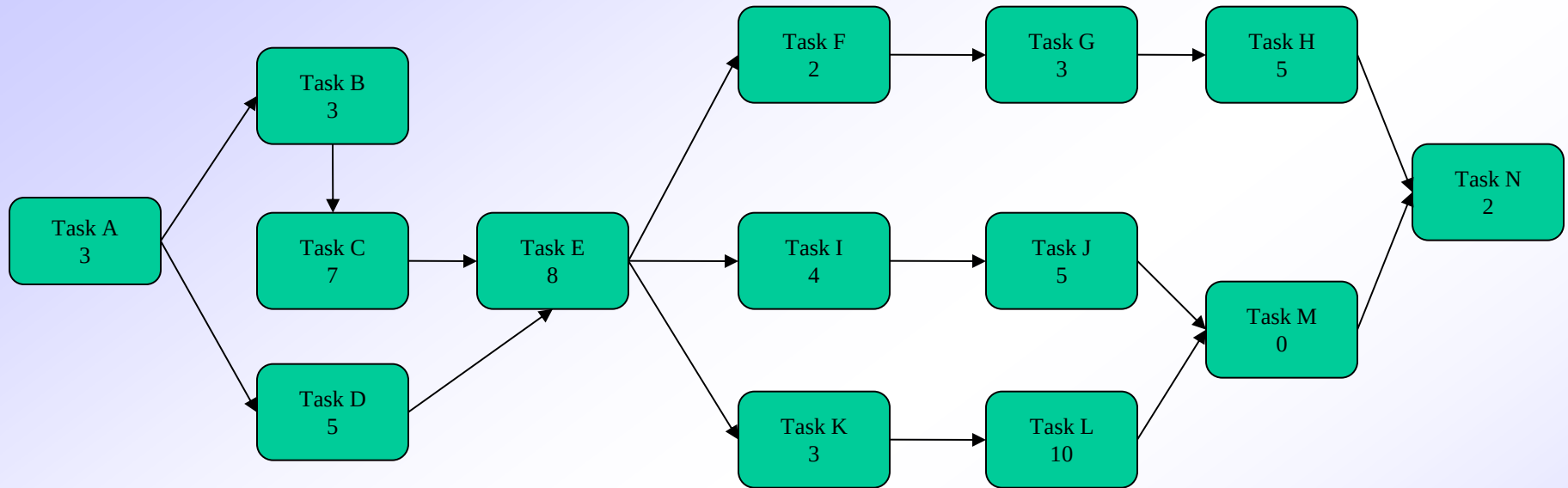
Purpose of a Task Network

- Also called an **activity network**
- It is a graphic representation of the task flow for a project
- It depicts task length, sequence, concurrency, and dependency
- Points out inter-task dependencies to help the manager ensure continuous progress toward project completion
- The critical path
 - A single path leading from start to finish in a task network
 - It contains the sequence of tasks that must be completed on schedule if the project as a whole is to be completed on schedule
 - It also determines the minimum duration of the project

The critical path:

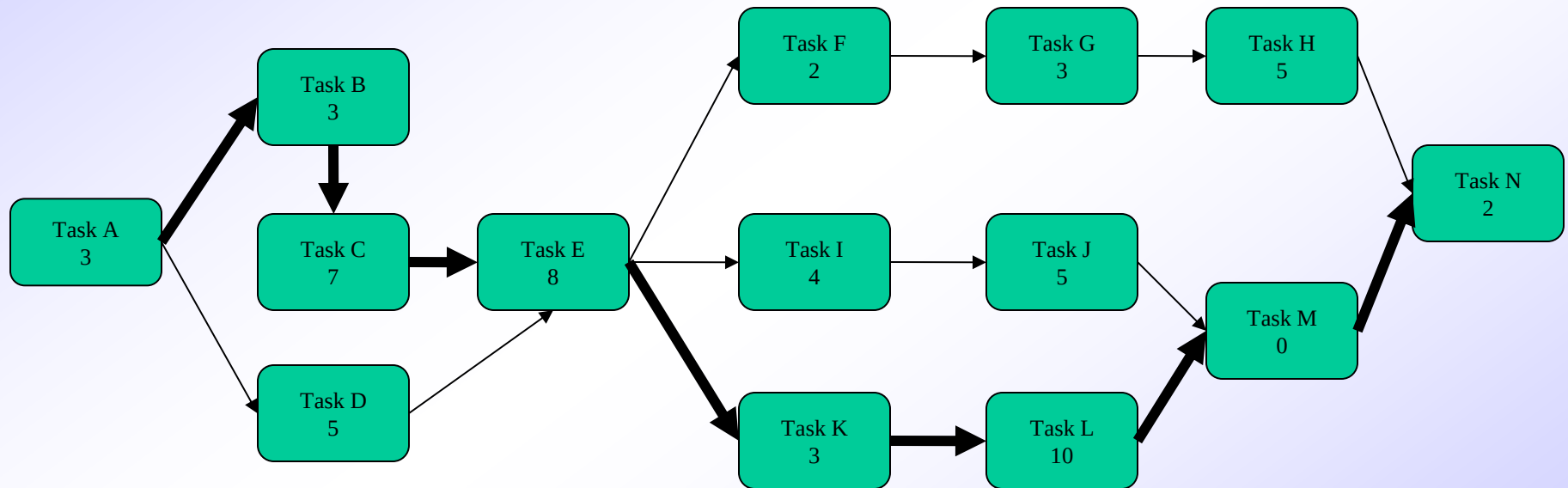
Longest sequence of activities in a project plan which must be completed on time for the project to complete on due date.

Example Task Network



Where is the critical path and what tasks are on it?

Example Task Network with Critical Path Marked



Critical path: A-B-C-E-K-L-M-N

Activity	Predecessor	Estimated Time (weeks)
A	None	5
B	A	3
C	B	1
D	B	10
E	C,D	2

Using the above data, i.e. activities, answer the following questions:

- 1) Draw a **Task network**.
- 2) Find all **paths to finish these** activities.
- 3) Find **critical path**.

Activity	Predecessor	Estimated Time (weeks)
A	None	5
B	A	3
C	B	1
D	B	10
E	C, D	2
F	B,E	3
G	F	4

Using the above data, i.e. activities, answer the following questions:

- 1) Draw a **Task network**.
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Review

Using the above data, i.e. activities, answer the following questions:

- 1) Draw a **Task network**.
- 2) Find all **paths to finish these** activities.
- 3) Find **critical path**.

Activity	Description	Predecessor	Estimated Time (Hours)
A	Drive home	None	0.5
B	Wash Clothes	A	4.0
C	Eat	B	0.5
D	Go to bank	A	1.0
E	Pay bill	D	0.5
F	Shopping	C,E	0.5
G	Drive home	F	0.5

Timeline Chart

Mechanics of a Timeline Chart

- **Also called a Gantt chart**; invented by Henry Gantt, industrial engineer, 1917
- **All project tasks** are listed in the far left column
- The next few columns may list the following for each task: **projected start date, projected stop date, projected duration**, actual start date, actual stop date, actual duration, **task inter-dependencies (i.e., predecessors)**
- To the far right are columns representing dates on a calendar
- The length of a horizontal bar on the calendar indicates the duration of the task
- **When multiple bars occur at the same time interval on the calendar, this implies task concurrency**
- A diamond in the calendar area of a specific task indicates that the task is a milestone; a milestone has a time duration of zero

Predecessor

						Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
Task #	Task Name	Duration	Start	Finish	Pred.										
1	Task A	2 months	1/1	2/28	None										
2	Milestone N	0	3/1	3/1	1										

CLASS EXERCISE

4/1 4/8 4/15 4/22 4/29 5/6 5/13 5/20 5/27 6/3

Task #	Task Name	Duration	Start	Finish	Pred.									
A	Establish increments	3	4/1		None									
B	Analyze Inc One	3			A									
C	Design Inc One	8			B									
D	Code Inc One	7			C									
E	Test Inc One	10			D									
F	Install Inc One	5			E									
G	Analyze Inc Two	7			A, B									
H	Design Inc Two	5			G									
I	Code Inc Two	4			H									
J	Test Inc Two	6			E, I									
K	Install Inc Two	2			J									
L	Close out project	2			F, K									

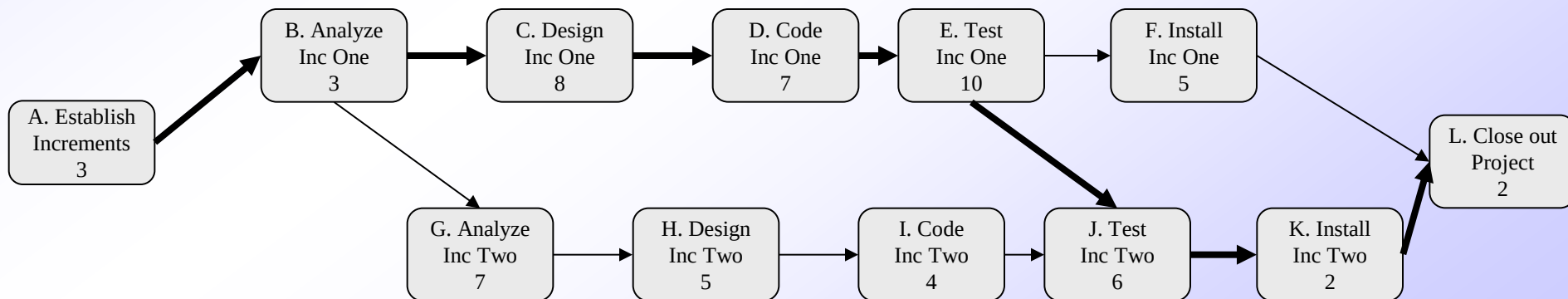
Using the above data, i.e. activities, answer the following questions:

- 1) Draw the **Gantt Chart**.
- 2) Find all **paths to finish these** activities.
- 3) Find **critical path**.
- 4) Draw a **Task network**.

SOLUTION

Timeline chart:

Task #	Task Name	Duration	Start	Finish	Pred.	4/1	4/8	4/15	4/22	4/29	5/6	5/13	5/20	5/27	6/3
A	Establish increments	3	4/1	4/3	None	■									
B	Analyze Inc One	3	4/4	4/6	A		■								
C	Design Inc One	8	4/7	4/14	B		■	■							
D	Code Inc One	7	4/15	4/21	C			■	■						
E	Test Inc One	10	4/22	5/1	D				■	■					
F	Install Inc One	5	5/2	5/6	E					■	■				
G	Analyze Inc Two	7	4/7	4/13	A, B		■	■							
H	Design Inc Two	5	4/14	4/18	G			■	■						
I	Code Inc Two	4	4/19	4/22	H				■	■					
J	Test Inc Two	6	5/2	5/7	E, I					■	■				
K	Install Inc Two	2	5/8	5/9	J						■	■			
L	Close out project	2	5/10	5/11	F, K							■	■		



Task network and the critical path: A-B-C-D-E-J-K-L

Task #	Task Name	Duration	Start	Finish	Pred.
A	Establish increments	3	4/1		None
B	Analyze Inc One	3			A
C	Design Inc One	8			B
D	Code Inc One	7			C
E	Test Inc One	10			D
F	Install Inc One	5			E
G	Analyze Inc Two	7			A, B
H	Design Inc Two	5			G
I	Code Inc Two	4			H
J	Test Inc Two	6			E, I
K	Install Inc Two	2			J
L	Close out project	2			F, K

Review

Activity	Description	Predecessor	Estimated Time (Hours)
A	Drive home	None	0.5
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F	Shopping	C,E	0.5
G	Drive home	F	0.5

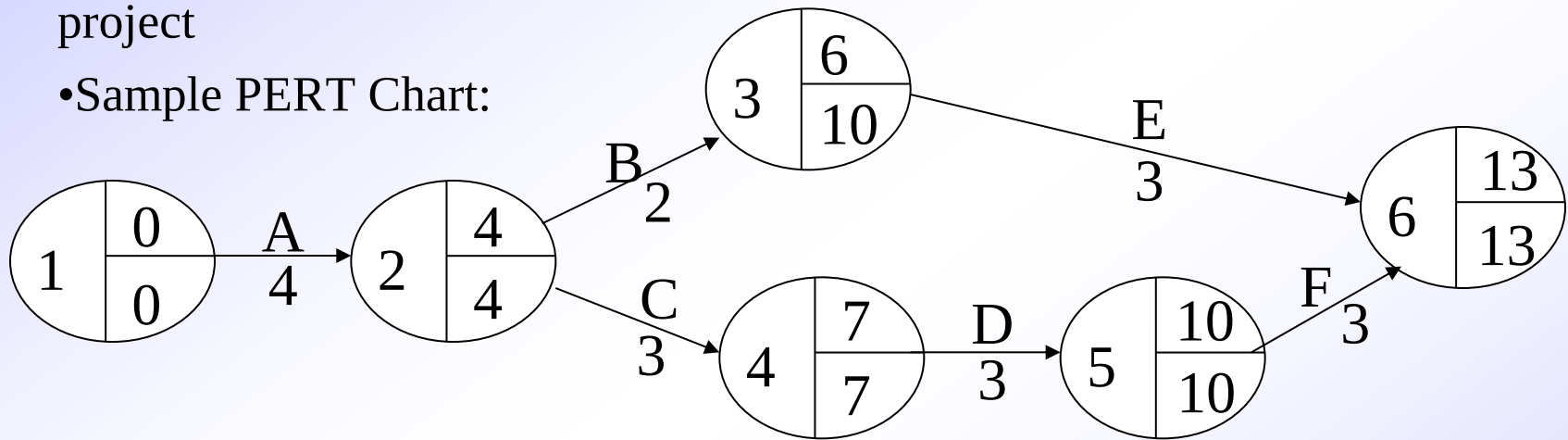
Using the above data, i.e. activities, answer the following questions:

- 1) Draw the **Gantt Chart**.
- 2) Find all **paths to finish these** activities.
- 3) Find **critical path**.
- 4) Draw a **Task network**.

PERT Charts

- PERT = Project Evaluation and Review Technique
- PERT chart = graphical representation of the scheduling of events in a project

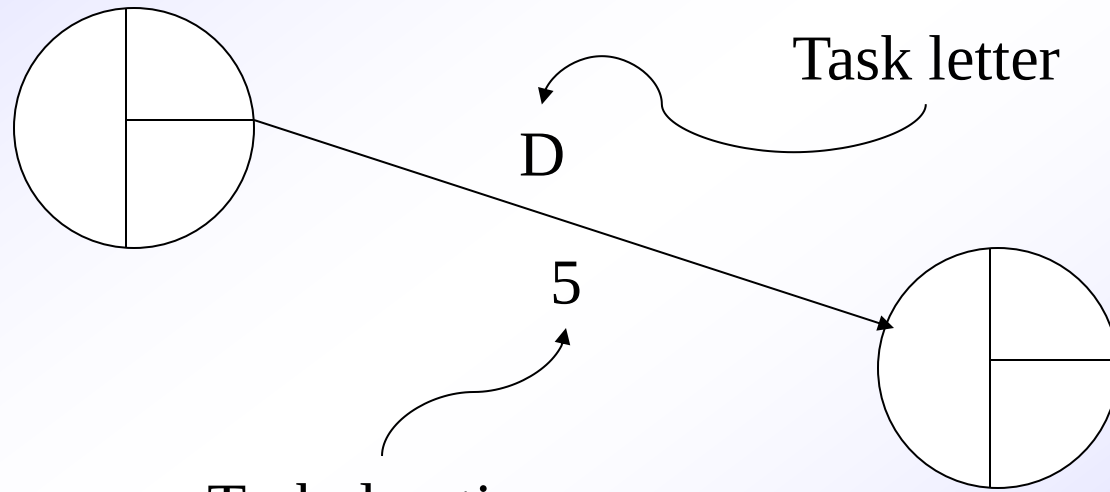
- Sample PERT Chart:



- A PERT chart is a graph
 - Edges are tasks/activities that need to be done
 - Nodes are the events or milestones
- Task edge T from event node E1 to event node E2 signifies:
 - Until event E1 happens, task T cannot be started
 - Until task T finishes, event E2 cannot happen
- Events often simply represent completion of tasks associated with arrows entering it

PERT Chart Task Edges

- Parts of a task/activity edge



- Task letter: Task duration

Often keyed to a legend to tell which task it represents

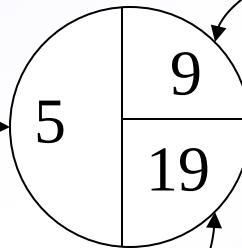
- Task duration = how long (e.g. days, hours) task will take

PERT Chart Event Nodes

Event Number:

Sequence number assigned

Only task edges indicate dependencies



Earliest Completion Time (ECT):

Earliest time this event can be achieved, given durations and dependencies

Latest Completion Time (LCT):

Latest time that this event could be safely achieved

Building a PERT Chart

Steps:

1. Make a list of all project tasks (and events if possible).
2. Find interrelated task dependencies (what task has to be completed before other tasks)
3. Draw initial PERT without durations, ECTs or LCTs
4. Estimate duration of each task
5. Fill in durations
6. Calculate ECTs and LCTs

•We will do this for an example system:

→ Generic software system with 3 modules

Dummy Tasks

Sometimes it is necessary to use *dummy tasks*:

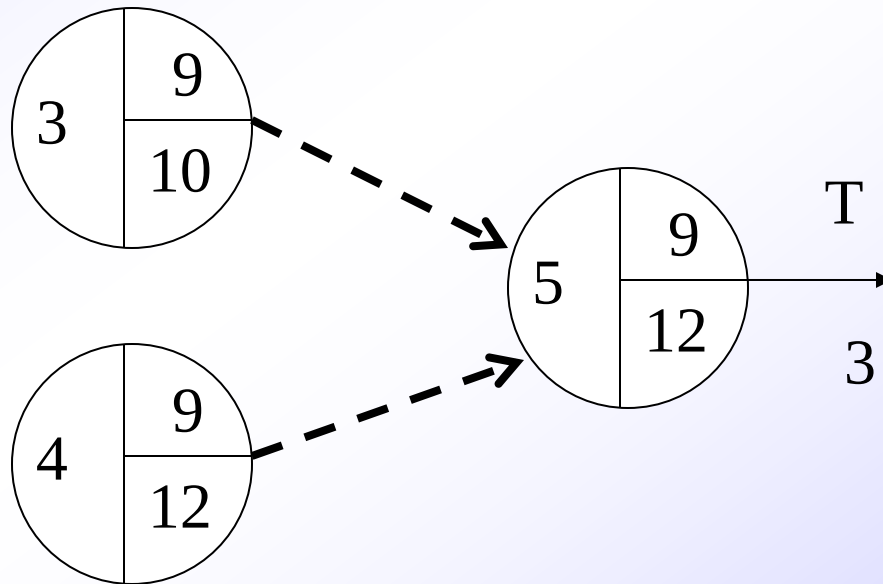
Shows the dependency between 2 events where no activity is performed

Example:

Events 3, 4 signify the compilation of separate modules.

Create an event 5 to signify “all modules compiled together”.

Denote dummy tasks using dash lines



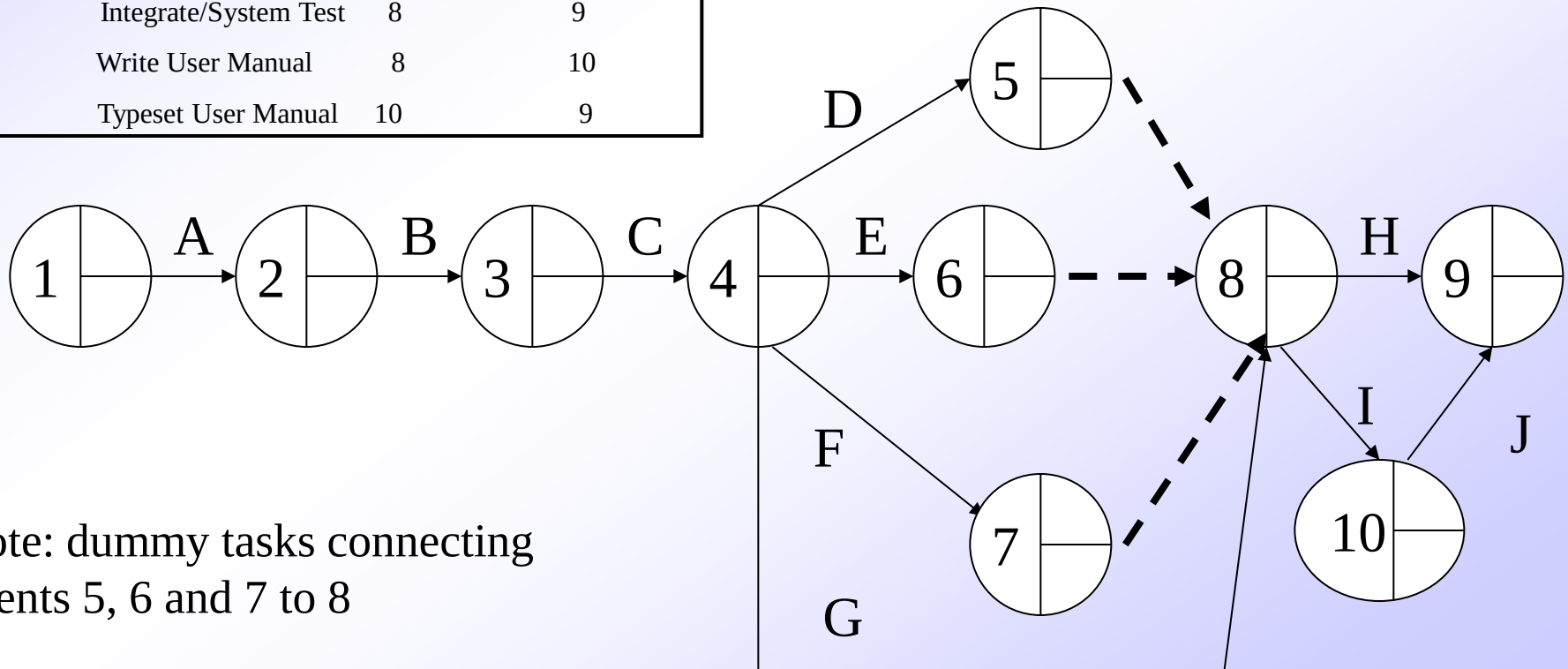
Example: Tasks with Dependencies

To start the PERT, identify the dependencies amongst tasks

TASK ID	Task Description	Preceed ID	Succ. ID	Estimated Time
A	Specification	1	2	3
B	High Level Design	2	3	2
C	Detailed Design	3	4	2
D	Code/Test Main	4	5	7
E	Code/Test DB	4	6	6
F	Code/Test UI	4	7	3
G	Write test plan	4	8	2
	Dummy Task	5	8	
	Dummy Task	6	8	
	Dummy Task	7	8	
H	Integrate/System Test	8	9	5
I	Write User Manual	8	10	2
J	Typeset User Manual	10	9	1

Software Example: Skeleton PERT Chart

TASK ID	Task Description	Preceed ID	Succ. ID
A	Specification	1	2
B	High Level Design	2	3
C	Detailed Design	3	4
D	Code/Test Main	4	5
E	Code/Test DB	4	6
F	Code/Test UI	4	7
G	Write test plan	4	8
	Dummy Task	5	8
	Dummy Task	6	8
	Dummy Task	7	8
H	Integrate/System Test	8	9
I	Write User Manual	8	10
J	Typeset User Manual	10	9



Estimating Durations

Suggestions for estimating durations of tasks:

- Don't just make up a number

- Look at previous similar tasks from other projects and use those as guidelines

- Try to identify factors such as difficulty, skill level

- Each weighting factor will help you make a better estimate

Factors to consider:

- Difficulty of task

- Size of team

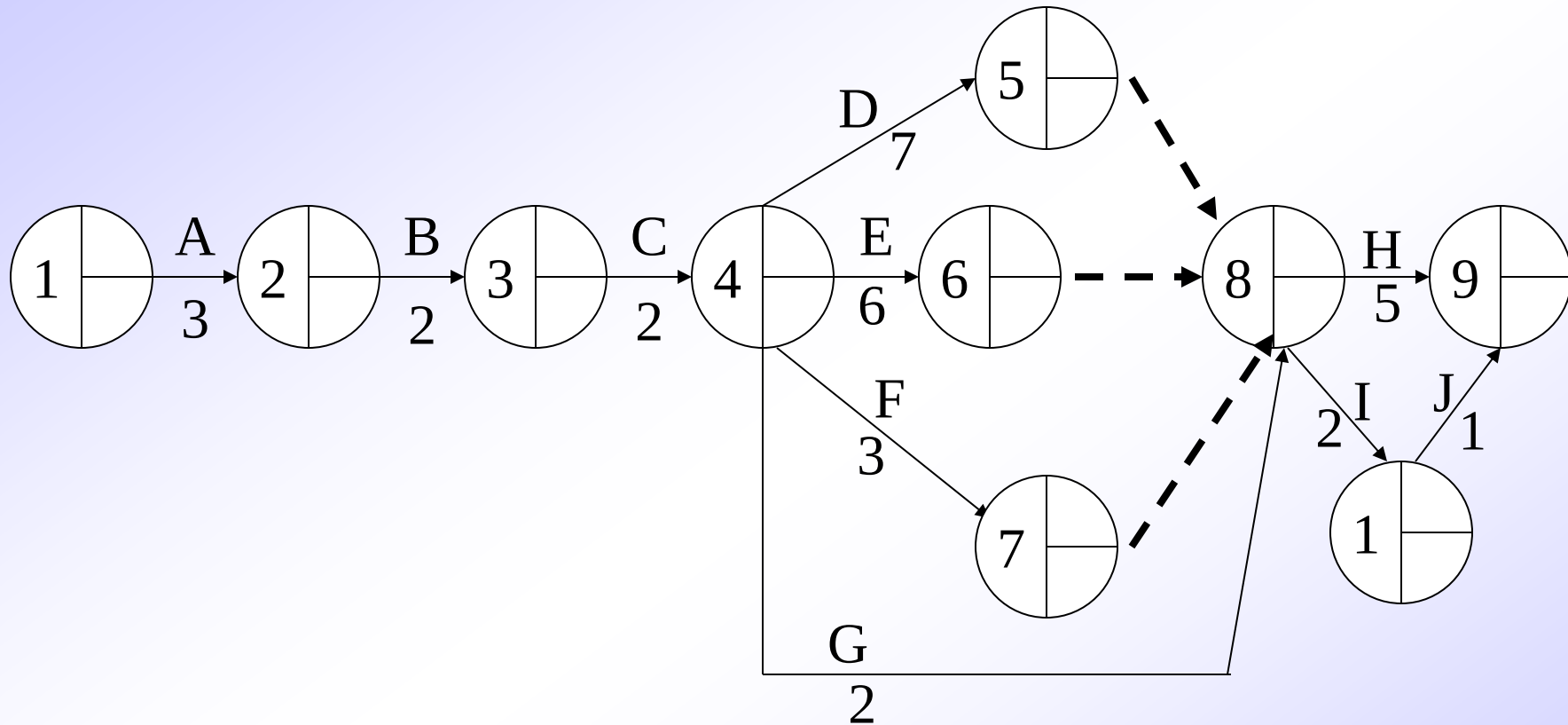
- Experience of team

- Number, attitude and availability of end users

- Management commitment

- Other projects in progress

PERT Chart With Durations



- Say we have estimated durations of all tasks (in days)
- New PERT chart, with durations filled in:
- Note, dummy tasks (dashed lines) always have a duration of zero

Calculating ECTs

ECT = earliest time event can be completed

To calculate:

For an event not depending on others: $ECT = 0$

Usually this is the first event

For an event E depending on one or more others:

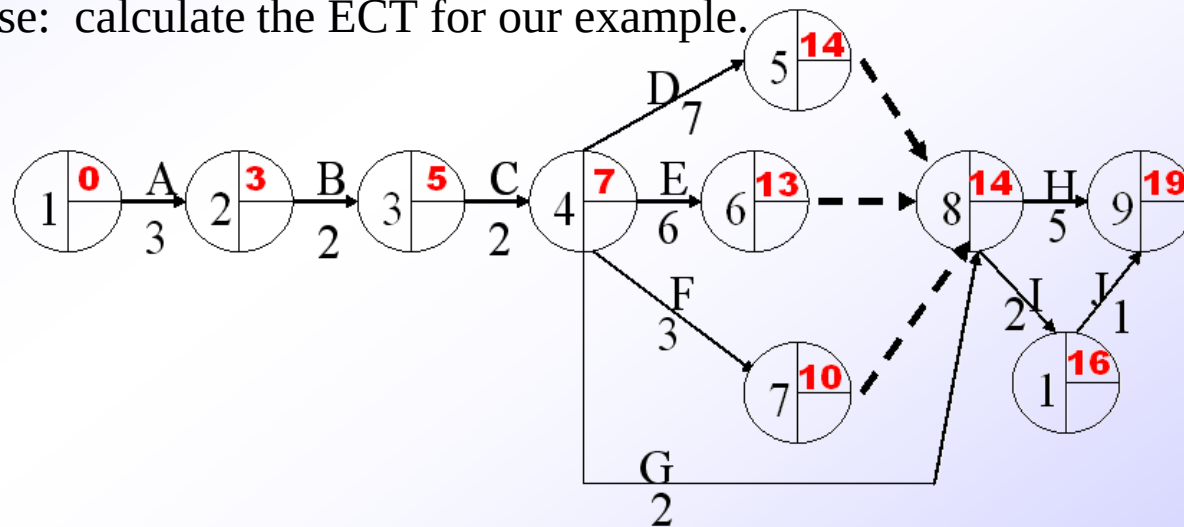
Calculate ECTs of event(s) that E depends on

Add duration(s) of task(s) leading to E

If E depends on more than one event, take MAX

Proceed left to right ($\rightarrow$) through the chart

Exercise: calculate the ECT for our example.



Calculating LCT

LCT = latest time event can be completed, while still finishing last task at indicated time

To calculate:

For an event which no other events depend on: $LCT = ECT$

Generally there will only be one such event

For an event E which one or more others depend on:

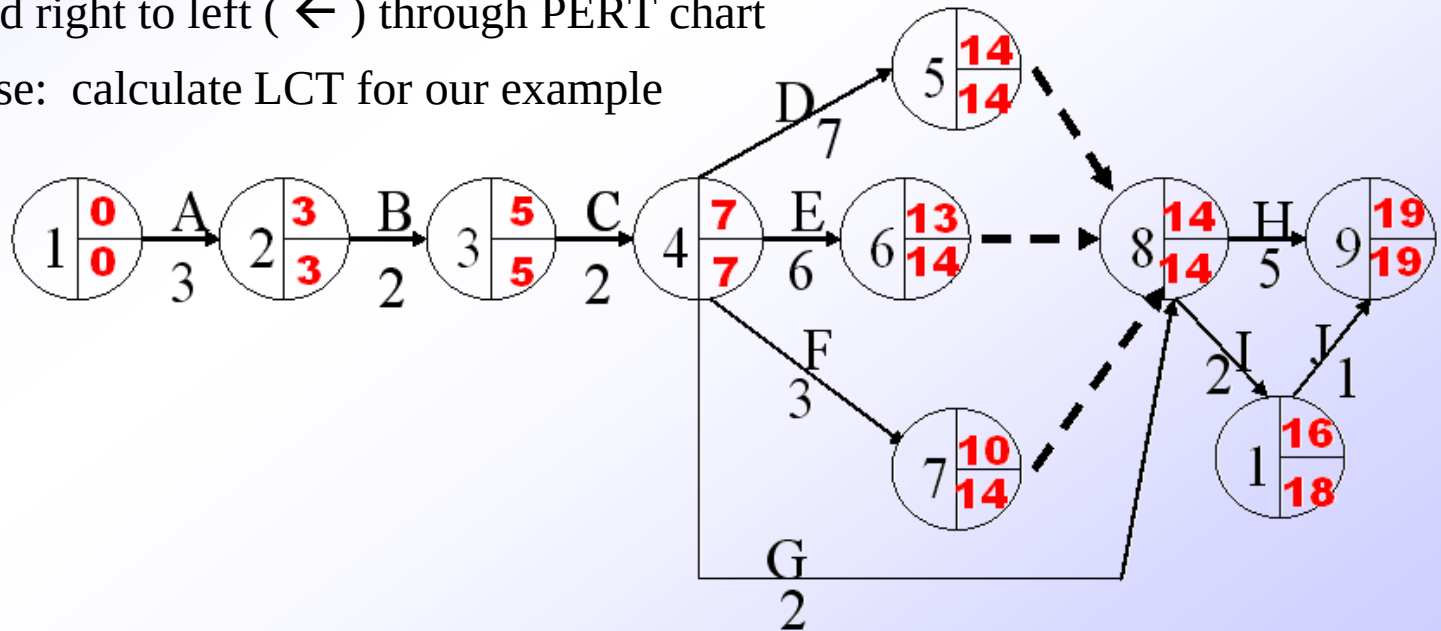
Calculate LCTs of event(s) that depend on E

Subtract duration(s) of task(s) leading from E

If more than one event depends on E, take MINIMUM

Proceed right to left ($\leftarrow$) through PERT chart

Exercise: calculate LCT for our example



Critical Path

